

Review

Car-Sharing Systems in Smart Cities: A Review of the Most Important Issues Related to the Functioning of the Systems in Light of the Scientific Research

Katarzyna Turoń 

Department of Road Transport, Faculty of Transport and Aviation Engineering, Silesian University of Technology, Krasińskiego 8 Street, 40-019 Katowice, Poland; katarzyna.turon@polsl.pl

Abstract: Currently, short-term car rental services (car-sharing) are a solution that is an alternative to individual motorization, which can be used in cities. With the level of sophistication of the relevant systems, there are more and more problems with their proper functioning. However, to understand the essence of some of the problems or properly introduce a new system to the market, it is important to understand how the systems work and the main relationships between the factors affecting car-sharing. This work aims to comprehensively present the most important aspects related to the functioning of car-sharing in light of the current research, as a form of compendium of knowledge about car-sharing. A review of the literature indicates that the leading areas of research in the field of car-sharing are seven topics related to the origin and history of services, the analysis of the functioning of the services market and criteria affecting the development of the relevant systems, the management of vehicle systems and location, issues related to the sustainable development of transport, IT systems, electromobility in car-sharing, and service optimization and modeling. A detailed analysis of the work carried out in this area allowed for the identification of 15 leading factors influencing car-sharing. This article supports the implementation and improvement of car-sharing services. In addition, it supports researchers during literature reviews and the search for a coherent list of the factors influencing car-sharing.



Citation: Turoń, K. Car-Sharing Systems in Smart Cities: A Review of the Most Important Issues Related to the Functioning of the Systems in Light of the Scientific Research. *Smart Cities* **2023**, *6*, 796–808. <https://doi.org/10.3390/smartcities6020038>

Academic Editor: Pierluigi Siano

Received: 25 January 2023

Revised: 21 February 2023

Accepted: 28 February 2023

Published: 3 March 2023



Copyright: © 2023 by the author. Licensee MDPI, Basel, Switzerland. This article is an open access article distributed under the terms and conditions of the Creative Commons Attribution (CC BY) license (<https://creativecommons.org/licenses/by/4.0/>).

Keywords: car-sharing services; car-sharing systems; car-sharing functioning; car-sharing operation; shared mobility; mobility management; smart cities; sustainable development

1. Introduction

Modern cities are developing at a very fast pace. Currently, 55% of the world's population lives in urban areas, and statistics indicate that this proportion is expected to increase to 68% by 2050 [1]. The phenomena of globalization and urbanization as well as the gradual shift of the population from rural areas are projected to add another 2.5 billion people to urban areas [2]. This type of increase will be particularly noticeable in cities with a population of less than 1 million [1], making the issue important both for small urban centers and for large agglomerations. The dynamic development of urban centers, apart from several advantages, is associated with many problems, including difficulties with one of the key factors of their economic development—transport and the elementary need of society, which is mobility [3]. Increasing the number of people staying in cities and at the same time meeting the related growing transport needs, regarding the fastest possible movement at any given time, is one of the challenges that urban transport managers face.

An attempt to reduce the phenomenon of increased traffic volume is also one of the aspects to be eliminated by the implementation of a sustainable transport development policy. According to its assumptions, modern transport should effectively meet the needs of society, be economically efficient, and at the same time have a positive impact on the environment, public health, economy, and urban planning [4]. Therefore, based on the guidelines of sustainable transport policy, the duties of cities include [5–8].

authors used the AHP method and derived data from the geographic information system (GIS) that was received from the system operator to identify potential future base stations of a car-sharing system operating in Shanghai. The authors relied on four main criteria, which included [66] potential users, demand for trips, destinations, and distance. This was based on 15 proposed evaluation indicators. The result of the investigation was to determine the type of station along with the number of parking spaces, between three and five, depending on the location in the city. In another work, based on the AHP method, the authors determined the profitability of the creation and maintenance of car-sharing stations in two districts in Shanghai [67]. The obtained results allowed to determine what actions the EVCARD operator should focus on to ensure the most efficient functioning of their system. Similar research, also based on the AHP method, was conducted in the French city of La Rochelle [68]. Apart from the indicated examples, no research studies were identified that would refer to the use of other types of multi-criteria decision support methods at the planning and implementing stages of car-sharing services. In addition, it was pointed out that most of the research on the location of charging stations or the appropriate relocation of vehicles is related to the optimization of car-sharing systems already existing on the market, rather than on any aspects of their implementation. It was also noted that the works relating to the modeling of systems are mainly based on the performance of simulations for the various types of data received from operators or traffic measurements. These data are mostly simulated, as they are computerized. Pagany, Camargo, and Dorne emphasized in their work [69] that case studies must be preceded by research that characterizes the detailed data or behavior patterns for a specific area.

4. Final Remarks

Car-sharing services are one of the alternative forms of mobility. Considering current trends, system development forecasts, and efforts to limit individual motorization in cities, car-sharing services may become a leading form of transport in cities. Particularly friendly to the development of car-sharing systems may be smart cities, which, thanks to the use of information and communication technologies to ensure the best possible interactivity and efficiency of urban infrastructure and its components, can offer residents properly functioning and accessible systems, while reducing many transport problems, for example, the social exclusion of transportation [70].

The conducted literature review showed that car-sharing systems have developed significantly in recent years. The leading factors of their development were the entertainment, infrastructure, and management aspects as well as the development of new technologies. However, for the systems to function better, it is important to take into account the many often-conflicting factors.

Therefore, for car-sharing operators or city authorities who want to implement or improve their car-sharing systems, it is recommended to pay attention to

- ensuring adequate (increased) availability of the fleet in bad weather conditions;
- properly considering the type of system that should be used around a given city, by considering the conflicting preferences for the use of round-trip versus free-floating car-sharing;
- applying an appropriate tariff of charges (by minute or kilometer);
- planning appropriate car-sharing zones that will make it possible to deploy one car-sharing vehicle per 1 square km of the zone's operating area, which will ensure high demand for services;
- introducing special benefits for using services on Mondays, e.g., additional credits for travel or the possibility of using a higher-class vehicle;
- making an appropriate selection of the vehicle fleet for the needs of the local community, by considering the availability of the infrastructure for electromobility;
- introducing special programs to promote the use of car-sharing by women to increase their interest in these services;
- striving to integrate car-sharing with public transport;

- providing special car-sharing zones near train stations, airports, and main communication points.

The fulfillment of the above criteria has a chance to translate into a better adaptation of car-sharing services for the needs of users and urban areas, which may have a positive impact on the development of the relevant systems and elimination of one of the problems of common mobility, i.e., the use of individual cars by drivers.

The conducted literature analysis showed that in terms of the functioning of car-sharing, researchers currently focus mainly on seven research areas, which include the origin of services, the analysis of the functioning of the services market and criteria affecting the development of the relevant systems, the management of vehicle systems and location, issues related to the sustainable development of transport, IT systems, electromobility in car-sharing, and service optimization and modeling. This is an important clue for researchers looking for new research gaps in the field of car-sharing.

In summary, through the analysis of the literature, the aim of the work was achieved in the form of indicating the most important aspects related to car-sharing in light of the current research. This work is a compilation of the existing knowledge about car-sharing and is a support for practitioners dealing with the subject of car-sharing, such as operators or municipal management, and a compendium of knowledge for researchers and people interested in shared mobility services.

The main research limitation of this work is the thematic limitation to only one database containing scientific works. In subsequent works, the author plans to carry out a literature review based on a larger number of databases and to compare the results.

Funding: This research received no external funding.

Data Availability Statement: The data presented in this study are available on request from the author.

Conflicts of Interest: The author declares no conflict of interest.

References

1. United Nations. Revision of World Urbanization Prospects. Available online: <https://population.un.org/wup/Publications/Files/WUP2018-Report.pdf> (accessed on 18 May 2022).
2. United Nations. Analysis and Policy Recommendations from the United Nations Secretary-General's High-Level Advisory Group on Sustainable Transport, Mobilizing Sustainable Transport for Development High-level Advisory Group on Sustainable Transport. 2016. Available online: <https://sustainabledevelopment.un.org/index.php?page=view&type=400&nr=2375&menu=1515> (accessed on 18 May 2022).
3. Thondoo, M.; Marquet, O.; Márquez, S.; Nieuwenhuijsen, M.J. Small Cities, Big Needs: Urban Transport Planning in Cities of Developing Countries. *J. Transp. Health* **2020**, *19*, 100944. [CrossRef]
4. Lai, C.M.T.; Cole, A. Measuring Progress of Smart Cities: Indexing the Smart City Indices. *Urban Gov.* **2022**, S2664328622000699, in press. [CrossRef]
5. Lejda, K.; Maździł, M.; Siedlecka, S.; Zielińska, E. The future of public transport in light of solutions for sustainable transport development. *Scientific Journal of Silesian University of Technology. Ser. Transp.* **2017**, *95*, 97–108.
6. Mayo, F.L.; Maglasang, R.S.; Moridpour, S.; Taboada, E.B. Impact of Transport Policies to Commuter Safety in Urban Cities of a Developing Country: A Sustainability and System Perspective. *Case Stud. Transp. Policy* **2022**, *10*, 2138–2152. [CrossRef]
7. Wyszomirski, O. Zrównoważony rozwój transportu w miastach a jakość życia. *Transp. Miej. i Reg.* **2017**, *12*, 27–32.
8. Soto, J.J.; Cantillo, V.; Arellana, J. Market Segmentation for Incentivising Sustainable Transport Policies. *Transp. Res. Part D Transp. Environ.* **2021**, *99*, 103013. [CrossRef]
9. Burkhardt, J. Limitations of mass transportation and individual vehicle systems for older persons. In *Mobility and Transportation in the Elderly*; Schaie, K.W., Pietrucha, M., Eds.; Springer: Berlin/Heidelberg, Germany, 2000; pp. 97–124.
10. Cheng-Min, F. New prospects of transportation mobility. *IATSS Res.* **2014**, *38*, 22–26.
11. Okraszewska, R.; Romanowska, A.; Wołek, M.; Oskarbski, J.; Birr, K.; Jamroz, K. Integration of a multilevel transport system model into sustainable urban mobility planning. *Sustainability* **2018**, *10*, 479. [CrossRef]
12. Jimenez, J. Smart Transportation Systems. In *Smart Cities*; McClellan, S., Jimenez, J., Koutitas, G., Eds.; Springer: Cham, Switzerland, 2018; pp. 123–133.
13. Benevolo, C.; Dameri, R.; D'Auria, B. Smart Mobility in Smart City. In *Empowering Organizations. Lecture Notes in Information Systems and Organisation*; Torre, T., Braccini, A., Spinelli, R., Eds.; Springer: Cham, Switzerland, 2016; pp. 13–28.
14. U.S. Department of Transportation. Smart City Challenge: Lessons for Building Cities of the Future. Available online: [https://cms.do.t.gov/sites/dot.gov/files/docs/Smart%](https://cms.do.t.gov/sites/dot.gov/files/docs/Smart%20City%20Challenge.pdf) (accessed on 11 June 2019).